

Finance Formula Sheet

Every formula from the Finance chapters (Ehrhardt & Brigham, Corporate Finance: Ch 2 Financial Statements & Cash Flows, Ch 3 Ratio Analysis, Ch 4 Time Value of Money), with what it is used for and how to apply it. Built for an open-note test: worked numbers come straight from the course's Computron example and TVM slides.

Ch 2 — Cash Flow & Performance Measures

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
Net cash flow $\approx$ Net income + Depreciation & amortization	Converting accounting profit to cash generated.	Add back the non-cash charges. Depreciation is usually the largest.
$NOPAT = EBIT \times (1 - T)$	Operating profit with financing effects removed.	Use EBIT, not net income. Computron 2023: $460 \times 0.75 = \$345$.
$NOWC = \text{Operating CA} - \text{Operating CL}$ Op. CA = cash + A/R + inventory (no ST investments) Op. CL = A/P + accruals (no notes payable)	Working capital tied up in running the business.	2023: $(50 + 520 + 820) - (400 + 240) = \750 .
Total net operating capital = $NOWC + \text{Net fixed assets}$	All investor-supplied capital used in operations.	2023: $750 + 3,500 = \$4,250$ (2022: \$3,480).
$FCF = NOPAT - (\text{Op. capital}_t - \text{Op. capital}_{t-1})$	Cash available to ALL investors after required investment; drives firm value.	2023: $345 - (4,250 - 3,480) = -\425 . Subtract the <i>change</i> in capital, not its level.
Operating profitability $OP = NOPAT \div \text{Sales}$ Capital requirement $CR = \text{Op. capital} \div \text{Sales}$	Decomposing ROIC into margin and capital intensity.	2023: $OP = 8.1\%$, $CR = 70.8\%$. Lower CR = better capital utilization.
$ROIC = NOPAT \div \text{Total net operating capital} = OP \div CR$	Best measure of operating performance; compare to WACC.	2023: $345 \div 4,250 = 8.1\% < WACC\ 10\% \rightarrow$ value destroyed.
$EVA = NOPAT - (WACC \times \text{Op. capital})$	Value created (or destroyed) in one year after charging for all capital, including equity.	2023: $345 - 0.10 \times 4,250 = -\80 . Positive = created value.
$MVA = \text{Market value of firm} - \text{Book value of capital}$ $\approx (\text{Shares} \times \text{Price}) - \text{Book equity}$ (if debt $\approx$ book)	Cumulative value created as judged by the market.	2023: $100 \times \$30 - 2,910 = \90 (2022: \$2,270).

Ch 2 — Tax Formulas

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT
$\text{NOL usable this year} = \text{MIN}(\text{cumulative unused NOL}, 80\% \times \text{taxable income})$	Applying loss carryforwards.	NOL \$72B, TI \$80B → min(72, 64) = \$64B; taxable income becomes \$16B; \$8B carries forward.
$\text{Max interest deduction} = 30\% \times \text{EBIT}$	Limit on deductible interest; excess carries forward 20 years.	EBIT \$50B, interest \$20B → deduct \$15B now, \$5B later.
$\text{Taxable dividends received} = (1 - 0.50) \times \text{Dividends}$	Corporate dividend exclusion (mitigates triple taxation).	\$28M received → \$14M taxable at 21%.
$\begin{aligned} \text{After-tax interest} &= \text{Interest} \times (1 - T) \\ \text{After-tax dividend} &= \text{Div} \times [1 - (1 - \text{excl.}) \times T] \end{aligned}$	Comparing corporate investments in bonds vs. preferred stock.	\$40M: 5.4% bond → \$1.62M after tax; 5.0% preferred → \$1.75M after tax (stock wins).
$\begin{aligned} \text{Average tax rate} &= \text{Total tax} \div \text{Taxable income} \\ \text{Marginal rate} &= \text{rate on the next dollar} \end{aligned}$	Personal tax analysis; decisions use the marginal rate.	Progressive system: average rate rises with income.
$\begin{aligned} \text{After-tax corporate bond yield} &= r \times (1 - T) \\ \text{Breakeven } T^* &= 1 - (\text{Muni yield} \div \text{Corporate yield}) \end{aligned}$	Muni vs. taxable bond choice.	6.4% corp, 5.0% muni, 24% bracket → 4.86% after tax < 5.0% → muni. $T^* = 21.9\%$.

Ch 3 — Ratio Analysis (all Computron 2024E)

RATIO	FORMULA	2024E VALUE · WHAT IT TELLS YOU
Liquidity — can it pay bills due within a year?		
Current ratio	Current assets ÷ Current liabilities	1,300 ÷ 700 = 1.9 (ind. 2.5)
Quick (acid-test) ratio	(Current assets – Inventories) ÷ Current liabilities	(1,300 – 660) ÷ 700 = 0.9 (ind. 1.3). Inventory is least liquid.
Asset management — how hard do assets work?		
Inventory turnover	COGS ÷ Inventories	5,580 ÷ 660 = 8.5× (ind. 9.0). COGS includes depreciation.
Days sales outstanding	Receivables ÷ (Sales ÷ 365)	530 ÷ 18.08 = 29.3 days (ind. 28). Days to collect.
Fixed assets turnover	Sales ÷ Net fixed assets	6,600 ÷ 3,700 = 1.8× (ind. 3.0)
Total assets turnover	Sales ÷ Total assets	6,600 ÷ 5,000 = 1.3× (ind. 1.5)
Debt management — how much leverage, and can it be serviced?		
Debt ratio (debt-to-assets)	Total debt ÷ Total assets (debt = notes payable + LT debt)	1,200 ÷ 5,000 = 24% (ind. 15%)
Debt-to-equity	Total debt ÷ Common equity	1,200 ÷ 3,200 = 0.38 (ind. 0.22)
Liabilities-to-assets	Total liabilities ÷ Total assets	1,800 ÷ 5,000 = 36% (ind. 30%)
Equity multiplier	Total assets ÷ Common equity	5,000 ÷ 3,200 = 1.56 (ind. 1.5). DuPont leverage term.
Times interest earned	EBIT ÷ Interest expense	620 ÷ 100 = 6.2× (ind. 13.0)
EBITDA coverage	(EBIT + Depr. & amort. + Lease pmts) ÷ (Interest + Lease pmts + Principal pmts)	(620 + 370 + 20) ÷ (100 + 20 + 0) = 8.4× (ind. 17.2)
Profitability — return on sales and on capital		
Net profit margin	Net income ÷ Sales	390 ÷ 6,600 = 5.9% (ind. 7.2%)
Operating profit margin	EBIT ÷ Sales	620 ÷ 6,600 = 9.4% (ind. 10.4%)
Basic earning power	EBIT ÷ Total assets	620 ÷ 5,000 = 12.4% (ind. 15.6%). Before taxes & leverage.

Return on assets	$\text{Net income} \div \text{Total assets}$	$390 \div 5,000 = 7.8\%$ (ind. 10.8%)
Return on equity	$\text{Net income} \div \text{Common equity}$	$390 \div 3,200 = 12.2\%$ (ind. 15.4%)
Market value — what does the market think?		
Price/earnings	$\text{Price per share} \div \text{EPS}$	$49 \div 3.90 = 12.6$ (ind. 16.8). \$ paid per \$1 of earnings.
Price/cash flow	$\text{Price} \div (\text{Net income} + \text{Depreciation}) \text{ per share}$	\$ paid per \$1 of cash flow.
Book value per share	$\text{Common equity} \div \text{Shares outstanding}$	$3,200 \div 100 = \$32.00$
Market/book	$\text{Price per share} \div \text{Book value per share}$	$49 \div 32 = 1.53$ (ind. 2.6)
Putting it together		
DuPont equation	$\begin{aligned} \text{ROE} &= \text{Profit margin} \times \text{Total assets turnover} \times \text{Equity multiplier} \\ &= (\text{NI} \div \text{Sales}) \times (\text{Sales} \div \text{TA}) \times (\text{TA} \div \text{Equity}) = \text{ROA} \times \text{EM} \end{aligned}$	2024E: $5.9\% \times 1.320 \times 1.563 = 12.2\%$. Levers: expense control, asset use, leverage.
Common-size statements	$\text{Balance sheet item} \div \text{Total assets} \cdot \text{Income item} \div \text{Sales}$	Removes size for peer comparison.
Percent change	$(\text{Value}_t - \text{Value}_{\text{base}}) \div \text{Value}_{\text{base}}$	Cumulative growth from the base year.

Ch 4 — Time Value of Money

FORMULA	WHAT IT IS USED FOR	HOW TO USE IT (EXAMPLES AT 10%)
$FV_n = PV \times (1 + I)^N$	Compounding a lump sum forward.	\$100, 3 yrs → $100 \times 1.331 = \$133.10$. Excel =FV(I, N, 0, -PV).
$PV = FV_n \div (1 + I)^N$	Discounting a future lump sum to today.	\$100 in 3 yrs → $100 \div 1.331 = \$75.13$. Excel =PV(I, N, 0, FV).
$N = \ln(FV \div PV) \div \ln(1 + I)$	How long to reach a target (e.g., doubling).	Double at 20%: $\ln 2 \div \ln 1.2 = 3.8$ yrs. Excel =NPER(I, 0, -PV, FV).
$I = (FV \div PV)^{(1/N)} - 1$	Implied growth or return rate.	Double in 3 yrs: $2^{(1/3)} - 1 = 25.99\%$. Excel =RATE(N, 0, -PV, FV).
$FVA_n \text{ (ordinary)} = PMT \times [((1 + I)^N - 1) \div I]$	Future value of equal end-of-period payments (savings plans).	\$100 × 3 yrs → $100 \times 3.31 = \$331.00$. Excel =FV(I, N, -PMT, 0).
$PVA_n \text{ (ordinary)} = PMT \times [1/I - 1/(I \times (1 + I)^N)]$	Present value of equal end-of-period payments (loan values, pensions).	\$100 × 3 yrs → \$248.69. Excel =PV(I, N, PMT, 0).
Annuity due = Ordinary annuity × $(1 + I)$ (PV or FV)	Payments at the beginning of each period (leases, rent).	PV $248.69 \times 1.1 = \$273.55$; FV $331 \times 1.1 = \$364.10$. Calculator BEGIN mode; Excel type = 1.
PV of perpetuity = $PMT \div I$	Payments that never end (consols, preferred stock).	\$100/yr at 10% → \$1,000. Use the periodic rate matching payment frequency.
PV of uneven stream = $\sum CF_t \div (1 + I)^t$	Any irregular cash flows; NPV of a project.	100, 300, 300, -50 → \$530.09. Excel =NPV(I, range) assumes first flow at t = 1.
$I_{PER} = I_{NOM} \div M$ · Periods = $N \times M$	Converting a quoted annual rate for intra-year compounding. Use for ALL calculations.	6% monthly → 0.5%/month; 5 yrs → 60 periods. Never calculate with the nominal rate directly.
$FV_n = PV \times (1 + I_{NOM} \div M)^{(M \times N)}$	Lump-sum growth with M compounding periods per year.	\$100, 12% semiannual, 5 yrs → $100 \times 1.06^{10} = \$179.08$ (daily: \$182.19).
$EFF\% \text{ (EAR)} = (1 + I_{NOM} \div M)^M - 1$	Comparing investments/loans with different compounding frequencies.	12% semiannual → 12.36%; quarterly 12.55%; monthly 12.68%; daily 12.75%. EFF% = nominal only if M = 1.
$APR = I_{PER} \text{ (from actual cash flows incl. fees)} \times M$ Modified APR = $(1 + I_{PER})^M - 1$	Comparing loans with different fees (Truth in Lending).	\$3,521.15 net of fee, $48 \times \$90 \rightarrow 0.8675\%/mo \rightarrow APR 10.41\%$; modified 10.92%.
Loan payment: solve $PVA_n = \text{Loan}$ for PMT $Interest_t = \text{Beginning balance}_t \times I$ $Principal_t = PMT - Interest_t$ $\text{Ending balance}_t = \text{Beginning balance}_t - Principal_t$	Building an amortization schedule.	\$1,000, 10%, 3 yrs → PMT \$402.11; yr-1 interest \$100, principal \$302.11, balance \$697.89; ends at \$0. Total interest \$206.34.
Fractional periods: $FV = PV \times (1 + I_{PER})^{(\text{days})}$	Odd holding periods with daily compounding.	$11.33463\%/365 = 0.031054\%/day$; 273 days → \$108.85.

Calculator checklist: clear TVM registers · $P/YR = 1$ · END mode (BEGIN for annuity due) · enter the unused variable as 0 (PMT = 0 for lump sums; PV = 0 or FV = 0 for annuities) · make one cash flow negative · use the periodic rate and total periods.